

HS 616: Managerial Thinking and Decision Making

Course Code: HS 616

Course Name: Managerial Thinking and Decision Making

L-T-P-C: 3-0-0-3

Prerequisites: IC 210 Probability, Statistics and Random Processes; or, with instructors permission

Intended for: B. Tech./M.S./Ph.D. students

Distribution: SHSS elective for B. Tech. students; A core or elective course for M.S./Ph.D. students.

Approval: 8th Senate

Course Contents

- **Introduction to decision making: descriptive, normative, prescriptive styles** Introduction to decision making; Decision-making approaches: Descriptive (psychological), Normative (rational), and Prescriptive (pragmatic); Bounded rationality and satisficing. [6 Lectures]
- **Introduction to decision analysis and problem framing** Study of decision analysis and technical tools for analyzing decisions; framing decisions, applications of decisions framing to marketing and management. [6 Lectures]
- **Decision analyses for certain and uncertain (probabilistic) decision situations** Methods of decision-making under certainty; Methods of decision-making under risk and uncertainty; Sequential decisions and decision trees; Multi-Criteria decision analysis and methods of resolving tradeoffs and conflicting objectives. [8 Lectures]
- **Prediction, forecasting, and judgments** Analytical methods for predictions; Anticipating and Forecasting using non-statistical methods; Role of Intuition versus Analysis in Judgments; Use of statistical (linear regression) models to capture human intuition; Judgment of Association and Causation; Counterfactual thinking. [8 Lectures]
- **Biases and heuristics in decision making** Introduction to different heuristics and biases, endowment effect, loss aversion, status-quo bias, inter-temporal biases, availability, representativeness, anchoring-and-adjustment, illusion of control, overconfidence, and confirmation bias. [8 Lectures]
- **Decision making in groups** From individual decision making to group decision processes; group polarization; groupthink; cognitive repairs; nudges; brainstorming; decision rules. [6 Lectures]

Textbooks:

1. J. Edward Russo & Paul Schoemaker, Winning Decisions: Getting it Right the First Time, Doubleday, 2002.
2. John S. Hammond, Ralph L. Keeney, & Howard Raiffa, Smart Choices: A Practical Guide to Making Better Decisions, Harvard Business School Press, 1999.

Reference Books:

1. Allen, David (2009). Making it all Work: Winning at the Game of Work and the Business of Life. New York: Penguin (ISBN-10: 0143116622).

2. Ayres, I. (2007). Super Crunchers: Why Thinking-by-Numbers is the New Way to Be Smart. New York: Bantam Books (ISBN 0553384732).
3. Bazerman, M.H., & Moore, D. (2005, 7th ed). Judgment in Managerial Decision Making. New York: Wiley (ISBN-13: 978-0-470-04945-7).
4. Gigerenzer, G. (2000). Adaptive Thinking: Rationality in the Real World. New York: Oxford University Press.
5. Gigerenzer, G., Todd, P., & ABC Research Group. (1999). Simple heuristics that make us smart. New York: Oxford University Press.
6. Gladwell, M. (2005). Blink: The Power Of Thinking Without Thinking. New York: Back Bay Books (ISBN-10: 0316010669).
7. Hardman, D. (2009). Judgment and Decision Making: Psychological Perspectives. New York: Wiley (ISBN: 978-1-4051-2398-3)
8. Heath, C. & Heath, D. (2007). Made To Stick: Why Some Ideas Survive And Others Die. New York: Random House (ISBN: 10-1400064287).
9. Kahneman, D. (2011). Thinking, fast and slow. New York: Farrar, Straus and Giroux.
10. Kahneman D., Slovic P., and Tversky, A. (Eds.) (1982) Judgment Under Uncertainty: Heuristics and Biases. New York: Cambridge University Press
11. Slovic, P. (2000). The Perception of Risk. Earthscan Publications.
12. Surowiecki, J. (2004). The Wisdom of Crowds. New York: Doubleday (ISBN 0-34-911605-9)

Articles:

Some journal articles on JSTOR and cases, articles, and teaching notes from Harvard Business School Press (<http://harvardbusinessonline.hbsp.harvard.edu>).